

AAU 2K11/19 Module 1 Questions

(Includes remembered & repeated questions from previous years exams)

1. Which one of the ff is TRUE about blood pressure regulation? (physiology)

- A. Stress relaxation increases blood pressure
- B. Baroreflex responses regulate blood pressures at intermediate time
- C. The sympathetic outflow is reduced to decrease blood pressure
- D. Parasympathetic nerve effects on heart is bigger to increase the blood pressure
- E. Blood pressure is regulated with the positive feedback homeostasis control system

ANS. C

Blood pressure regulation

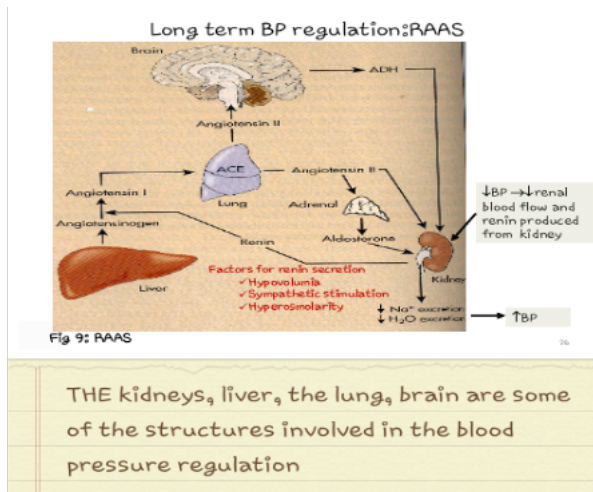
- Short-term
 - Baro-reflex
 - Vagus and glossopharyngeal nerve
 - Receptor adaptation; reduced pressure sensitivity
 - Maladaptation
- Intermediate-term
 - Fluid shift
 - Stress relaxation
- Long-term
 - RAAS

On physiology slide, B can't be the answer.

Examples of Negative Feedback control

- Blood Sugar Regulation
- Osmoregulation
- Control of Temperature
- Plasma CO₂, pH regulation
- Regulation of blood volume and pressure
- Endocrine gland secretion
- Regulation of acute haemorrhage
- End-product inhibition
- Pre-synaptic inhibition

E can't be the answer, because Blood pressure is regulated with the Negative feedback homeostasis control system, Not by the positive



Read into this picture bedemb, there were other questions related to this system(RAAS renin-angiotensin-aldosterone system). On physiology slide.

The **sympathetic** and **parasympathetic** branches of the **autonomic nervous system** have opposing effects on blood pressure. Sympathetic activation leads to an elevation of **total peripheral resistance** and **cardiac output** via increased **contractility** of the heart, **heart rate**, and arterial **vasoconstriction**, which tends to increase blood pressure. Conversely, **parasympathetic** activation leads to decreased **cardiac output** via decrease in heart rate, resulting in a tendency to lower blood pressure.

By coupling **sympathetic** inhibition and **parasympathetic** activation, the baroreflex maximizes blood pressure reduction. **Sympathetic** inhibition leads to a drop in peripheral resistance, while parasympathetic activation leads to a depressed heart rate (**reflex bradycardia**) and **contractility**. The combined effects will dramatically decrease blood pressure. In a similar manner, **sympathetic** activation with **parasympathetic** inhibition allows the baroreflex to elevate blood pressure.

Regulation of Arterial Blood Pressure. Several systems contribute to the regulation of arterial blood pressure. One of these, the baroreceptor system, is a simple and excellent example of a rapidly acting control mechanism. In the walls of the bifurcation region of the carotid arteries in the neck, and also in the arch of the aorta in the thorax, are many nerve receptors called baroreceptors, which are stimulated by stretch of the arterial wall. When the arterial pressure rises too high, the baroreceptors send barrages of nerve impulses to the medulla of the brain. Here these impulses inhibit the vasomotor center, which in turn decreases the number of impulses transmitted from the vasomotor center through the sympathetic nervous system to the heart and blood vessels. Lack of these impulses causes diminished pumping activity by the heart and also dilation of the peripheral blood vessels, allowing increased blood flow through the vessels. Both of these effects decrease the arterial pressure back toward normal. Conversely, a decrease in arterial pressure below normal relaxes the stretch receptors, allowing the vasomotor center to become more active than usual, thereby causing vasoconstriction and increased heart pumping, and raising arterial pressure back toward normal.

2. Which of the ff linkages CAN'T be digested by humans? (Biochemistry)

- A. Alpha (1,4) linkage
- B. Alpha (1,6) linkage
- C. Beta (1,4) linkage
- D....
- E....

ANS. C

Cellulose(made of Beta 1-4 glycosidic linkages between Glucose units) can't be digested by humans.....Humans lack cellulase enzyme.

3. About body fluid, calculation question meto neber. yesewyew mass is given. A 75 kg old man minamin, filtration rate calculate madreg. (physiology)

Body fluid: water

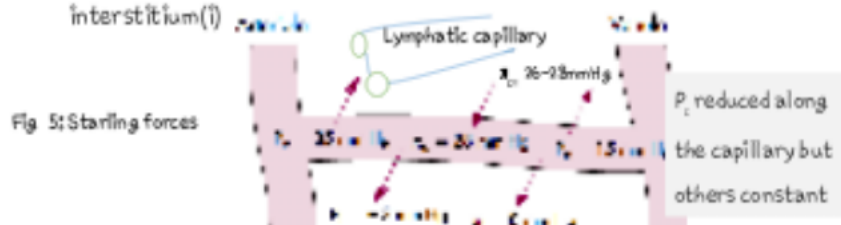
- The total body water is affected by
 - Age, sex and type of tissue (lean, fat(obesity))
 - Adult male; 60% of the total body weight
 - Adult female; 50% of the total body weight, 1kg mL
 - Amount of fat
 - Steroid hormones
- Can be quantified
 - Total body water = 60% * total body weight(kg) male
 - Total body water = 50% * total body weight(kg) female

Body Fluid Compartments

- Intracellular compartment (IC)
 - ICF or cytosol
 - 2/3 of TBW
- Extracellular compartment (EC) -
 - ECF
 - 1/3 of TBW
 - Found in other three compartments
 - In the interstitium; interstitial/tissue fluid (3/4 of ECF)
 - In the blood vessels; intravascular fluid (plasma); 3/14 of ECF
 - Special compartment; transcellular fluid (1/28 of ECF)
- The proportion is regulated otherwise, flaccid (cellular solute elevated) or turgidity of cell occur

The fluid exchange b/n compartments...

- Oncotic and hydrostatic pressure (starling forces)
- Filtration rate $= K_f [(P_c - P_i) - \sigma(\pi_c - \pi_i)]$
 - Where: K_f = filtration coefficient (vary capillary to capillary (K_f of kidney 100X higher than the skeletal muscle), P = hydrostatic pressure in the capillary and in the interstitium (i), σ reflection coefficient of proteins and π colloid oncotic pressure in the capillary (c) and interstitium (i)



On this slide, the formula for filtration rate;

Filtration rate = $K_f [(P_c - P_i) - \sigma(\pi_c - \pi_i)]$, is important. Yazut tiyakew mimelesew yehen formula bemetekem neber.

4. There was this question about Respiratory distress syndrome. (Biochemistry)

ANS. It's a disorder that occurs in infants characterized by the absence of dipalmitoyl lecithin. Also called dipalmitoyl phosphatidyl choline.

8. Act as surfactants (agents lowering surface tension).

- E.g. dipalmitoyl phosphatidylcholine is an important lung surfactant.
- **Respiratory distress syndrome** infants is associated with insufficient production of this surfactant.

5. Although Transfat has a similar structure to unsaturated fat, it's more dangerous than saturated fat. Why? (Biochemistry)

- A. Their ability to form Triglycerides C..... D....
 B. Their ability tominamin...Linear strcture E.....

Can't recall the other choices, try to find out the reason by urself! Gn zis might help

How are **trans fats** similar to **saturated fats**? Trans fats are **similar** in structure to **saturated fats**, and they also raise levels of total cholesterol **and** low density lipoprotein (LDL), or "bad," cholesterol. **Trans fats** may also raise blood triglycerides **and** lower high density lipoprotein (HDL), or "good," cholesterol.

*Focus on pros and cones of transfats.

Trans Fats – The Double Whammy

- Trans fats come from adding hydrogen to vegetable oil (unsaturated fats) through a process called **hydrogenation**.
- **"Partially hydrogenated"** oils may contain higher levels of trans fats.
- Trans fats are more solid than oils.
- Trans fats increase the shelf life of foods.
- Unlike other fats, trans fats **both** raise your "bad" (LDL) cholesterol **and** lower your "good" (HDL)



TRANS FATTY ACIDS

✓ Present in fast foods, dairy products & in hydrogenated edible oils.

✓ They are injurious to health.

✓ However, they are widely used in food industry as they increase the shelf life of the fried food.

✓ Fast food preparations have a high TFA content.



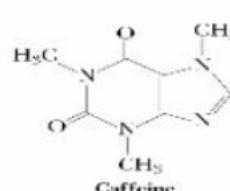
6. Which nucleoside blood vessel dilation yetekmal? (Biochemistry)

- A. Thymidine
- B. Adenosine
- C. Guanosine
- D. Uridine
- E. Cytidine
- ANs. B

Common

Adenosine: Exceptionally important autocoid

- Cause blood vessel dilation;
 - e.g. Exercising muscle release adenosine
- Regulation of Smooth muscle contraction
 - e.g. Blocking the electric flow from SA node to cardiac muscle
 - Important during *Supraventricular tachycardia*
 - Marketed as *AdenocardTM*
- Neurotransmitter release
- Metabolism of fat
- Promotes sleepiness
- Caffeine can reacts with extracellular adenosine



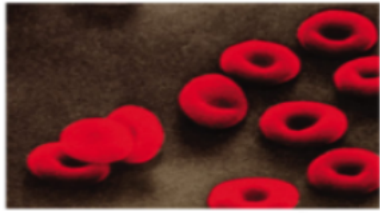
Caffeine

7. A question about Haemoglobin and sickle cell anaemia .

* Sickle cell mifeterew is when valine replaces Glutamic acid(glutamate). (Biochemistry)

Primary structure ...

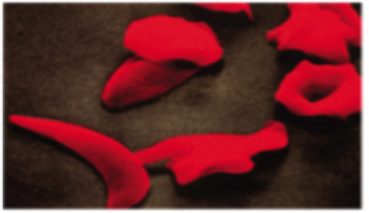
Change in amino acid sequence may cause problem!
Hemoglobin - 574 amino acids



10 µm

Val	His	Leu	Thr	Pro	Glu	Glu	...
1	2	3	4	5	6	7	

(a) Normal red blood cells and the primary structure of normal hemoglobin



10 µm

Val	His	Leu	Thr	Pro	Val	Glu	...
1	2	3	4	5	6	7	

(b) Sickled red blood cells and the primary structure of sickle-cell hemoglobin

glutamic acid is acidic & polar
valine is non-polar = tries to "hide" from water of cell by sticking to another hemoglobin molecules.

8. Which one of the ff Amino acid is both ketogenic and glucogenic?
(Biochemistry)

- A. Leucine
- B. Lysine
- C. Valine
- D. Proline
- E. Tryptophan

Ketogenic	Mixed Type	Glucogenic
Leu	Ile	Ala, Arg, Asp, Asn, Glu, Gln, Ser, Met, Pro, Gly, His, Thr, Val, Cys
Lys	Tyr	
	Trp	
	Phe	

ANS. E

This thing, amino acids being glucogenic, ketogenic or both, module 3 deres teketilon meto neber.

9. Which one of the ff is an Indispensable amino acid? (Biochemistry)

This question senefeten choices wust mels alneberewm. But

Amino Acid Classification Based on Nutritional Value

- ▶ Based upon whether the AAs can be synthesized in human body or not
 - ▶ Indispensable or essential AAs
 - Not synthesizable in the body in adequate amounts
 - Val, Ile, Thr, Trp, Arg, Leu, Lys, Met, Phe and His.
 - 'VITTAL LyMPH'
 - ▶ Dispensable or non-essential AAs
 - Can be synthesizable in the body from the essential ones
- Arg & His are semi-essential:

10. Which bond stabilizes the secondary structures of proteins? (Biochemistry)

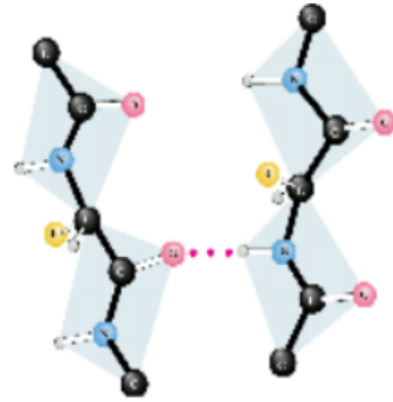
- A. Disulfide bridge
- B. Hydrophobic interaction
- C. Hydrogen bond
- D.
- E.

ANS. C



Secondary structure of proteins: Local folding

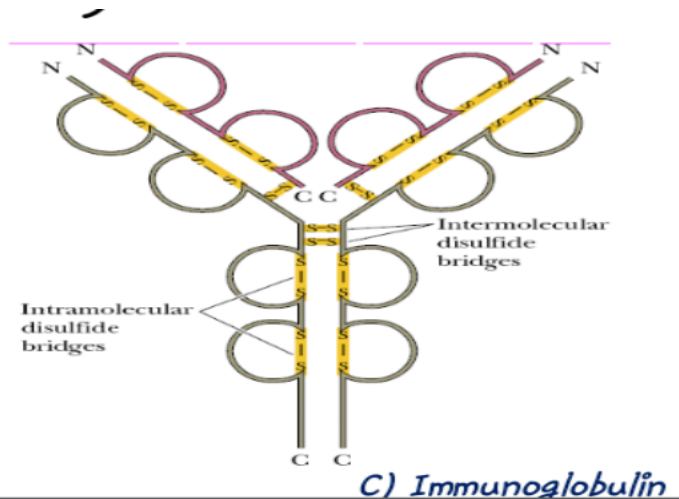
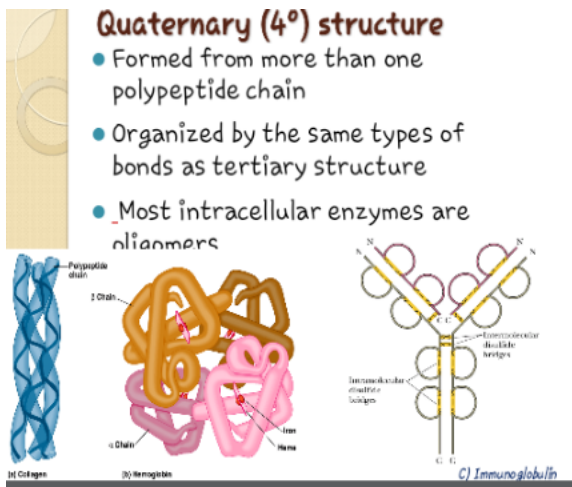
- Describes about folding pattern along short sections of polypeptide
- The Development of regular patterns of hydrogen bonding result in distinct folding patterns



11. The protein Immunoglobulin has

- A. Primary structure
- B. Secondary structure
- C. Tertiary structure
- D. Quaternary structure

ANS. D



12. There were two questions about loss of native form of proteins and heat shock proteins. Related things include chaperones and their contribution in protein folding. Also GroEL and GroES eneman endehonuna min endemiseru... (Biochemistry)

* Andu question which one is true bilo neber miteykew, and the answer was "GroEL is responsible for protein folding" milew.

13. Which protein is made from 3 subunits? (Biochemistry)

- A. Immunoglobulin
- B. Aldolase
- C. Alcohol dehydrogenase
- D. Glutathione reductase
- E. Alkaline phosphatase

ANS. B

Examples of Proteins with Quaternary Structure

Protein	Number of Subunits
Alcohol dehydrogenase	2
Immunoglobulin	4
Malate dehydrogenase	2
Superoxide dismutase	2
Triose phosphate isomerase	2
Glycogen phosphorylase	2
Alkaline phosphatase	2
6-Phosphogluconate dehydrogenase	2
Phosphoglucosomerase	2
Tyrosyl-tRNA synthetase	2
Glutathione reductase	2
Aldolase	3

14. Which of the ff can be acidic to the kidney?

- A. Acetylcholine B. (HCO₃)⁻ C. NH₄⁺ D. ... E. ...

ANS. C

15.is a visualization of all of an organism's chromosomes together, each labeled a different color. (Cytology)

ANS. Special karyotype

16. Which of the ff is true?

- A. Physiology studies about the structure of the body.
- B. Physiology studies the purpose of process at systemic level only.
- C. The scope of physiology is starting from microscopic structure to the organism level.
- D. The mechanistic approach of physiology studies the purpose of process in the body.
- E. The concept in physiology does not contribute for the current medicine.

ANS. C is the answer...A mayhnew is because it studies about the function of the body....D demo mayhnew Teleological approach new ke purpose ga miyayazew. B and E demo, they're obviously false.

17. A question about electrolyte regulation...Sodium level in the blood? (physiology)

ANS. 135-145mM/L

Vary according to the body fluid

Electrolyte	Plasma (mM/L)	CSF	Urine
Sodium	135-145	138-150	40-220
Potassium	3.5-5	0.35-3.5	25-125
Chloride	98-107	118-132	110-250
Bicarbonate	22-29	---	---
Calcium	2-15-2.55		Upto 7.49
Phosphate	0.8-1.45		12-42

18. Which of the ff is an immediate homeostatic response for the change made in the body?(Physiology)

- A. Cold induced enlargement of thyroid gland.
- B. Hot temperature induced increased sweat gland effectiveness
- C. Gene expression alteration for a different behavior to be observed
- D. Cold exposure induced surface vessels constriction
- E. Environment change induced acclimatization

ANS. D

19. Which of the ff is the most important chemical buffer of the plasma?(Physiology)

- A. $(\text{HCO}_3^-)/\text{H}_2\text{CO}_3$ C. Organic phosphate esters
 B. $(\text{HPO}_4)^{2-}/(\text{HPO}_4)^-$ D. Protein E. Hemoglobin

ANS. A(bicarbonate).....Intracellular sihon demo Phosphate

20. Which one of the ff is correct regarding chaperones?(Biochemistry)

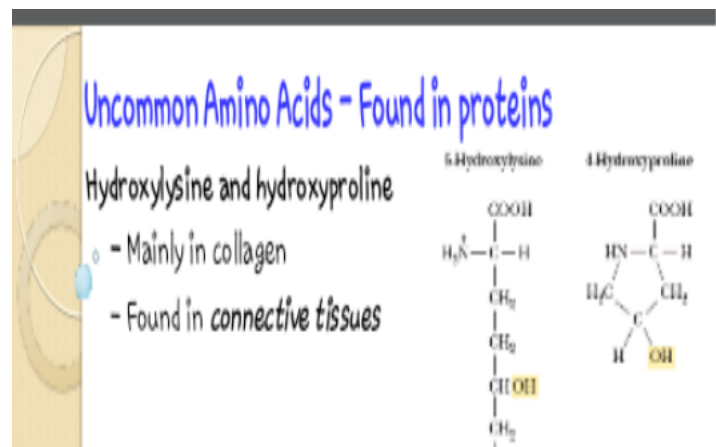
- A. They promote aggregation of non-native proteins.
 B. They are synthesized only when they are induced by a stimuli
 C. Usually recognize hydrophilic region of aminoacid residues
 D. Associated with the degraded proteins during folding process
 E. They promote premature folding of proteins

ANS. B(not sure)

21. Which of the ff aminoacids are hydroxylated during the formation of collagen protein?(Biochemistry)

- A. Cysteine and Methionine
 B. Proline and Lysine
 C. Lysine and Leucine
 D. Arginine and Glycine
 E. Leucine and Isoleucine

ANS. B



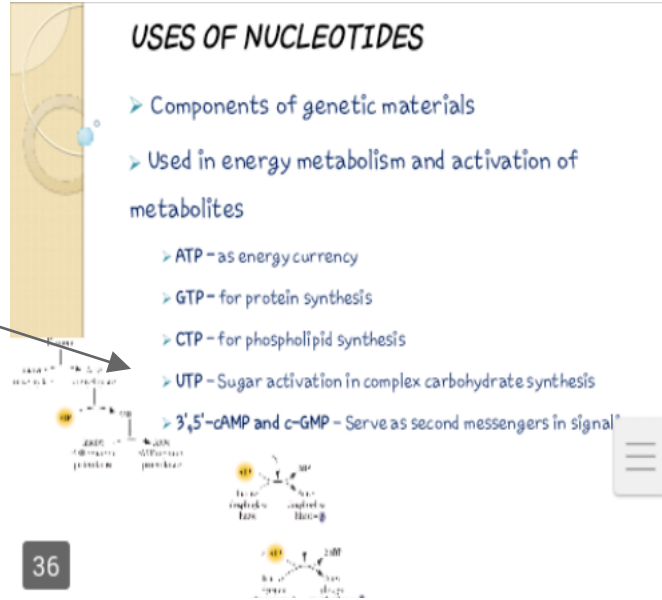
22. The nucleotide which is important for activation of monosaccharide units in polysaccharide synthesis is (Biochemistry)

- A. Uridine Triphosphate
- B. Cyclic Adenosine Monophosphate
- C. S-adenosyl methionine
- D....
- E...

ANS. A

USES OF NUCLEOTIDES

- Components of genetic materials
- Used in energy metabolism and activation of metabolites
 - ATP - as energy currency
 - GTP - for protein synthesis
 - CTP - for phospholipid synthesis
 - UTP - Sugar activation in complex carbohydrate synthesis
- 3',5'-cAMP and c-GMP - Serve as second messengers in signal



*Read yelelochunim function.

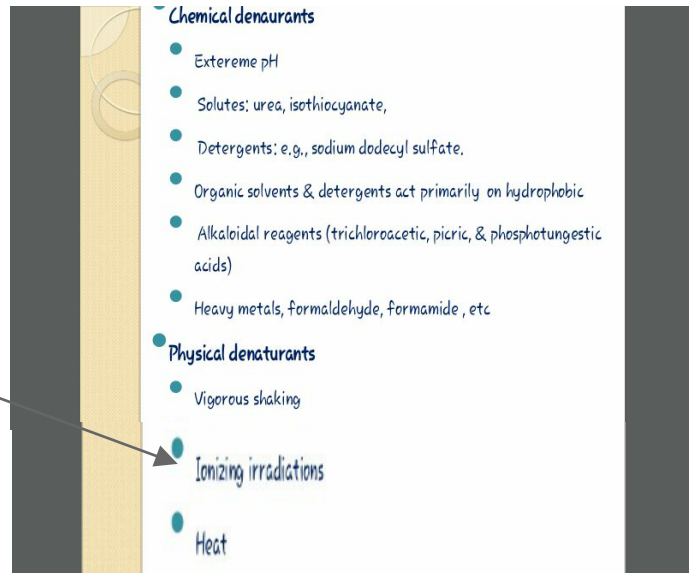
23. Which of the ff group of aminoacids is mainly located in the inner core of globular proteins?(Biochemistry)

- A. Glutamate, Arginine, Lysine
- B. Valine, Isoleucine, Glycine
- C. Aspartate, Serine, Methionine
- D. Histidine, Glutamine, Asparagine
- E. Serine, Tyrosine, Glutamate

ANS. YULLI: The inner core of globular proteins is hydrophobic so the answer is B.

24. Among the ff types of protein denaturing agents, which one is classified as physical denaturant?(Biochemistry)

- A. Extreme PH
- B. Ionization radiations
- C. Heavy metals
- D. Organic solvents
- E. Detergents



ANS. B

25. The most important types of amino acid residues in collagen formation?(Biochemistry)

- A. Glycine and Proline
- B. Lysine and Hydroxylysine
- C. Cysteine and Methionine
- D. Glycine and Lysine
- E. Alanine and Histidine

ANS. A

26. Centrioles self duplicate at which phase of the cell cycle?(Cytology)

- A. G0
- B. G1
- C. S-phase
- D....
- E.....

ANS. C,....They self-duplicate at S-phase,

They grow to maturity at G2 phase.

27. The branch points of amylopectin in starch granules are connected by (Biochemistry)

- A. Alpha (1,4) B. Beta (1,4) C. Alpha (1,6) D. Beta (1,6) E. Alpha (1,2)

ANS. C

28. Which of the ff is an example of heteropolysaccharide? (Biochemistry)

- A. Starch B. Glycogen C. Cellulose D. Chitin E. Hyaluronic acid

ANS. E

29. Which of the ff is not an organic biomolecule? (Biochemistry)

- A. Carbohydrates
B. Proteins
C. Nucleic acids
D. Lipids
E. Water

ANS. E

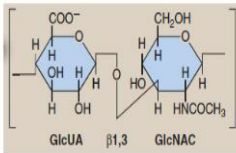
30. The vitreous humor of eye is composed of (Biochemistry)

- A. Heparin
- B. Chondroitin sulfate
- C. Keratan sulfate
- D. Hyaluronic acid
- E. Dermatan sulfate

ANS. D

Hyaluronic acid

- Disaccharide unit:
N-acetylglucosamine and glucuronic acid.
- unique among the GAGs in that it **does not contain any sulfate** and is **not found covalently** attached to proteins as a proteoglycan.
- It is, however, a component of non-covalently formed complexes with proteoglycans in the ECM.
- Only GAG** present both in animals and bacteria.
- Found in synovial fluid, vitreous humor, ECM of loose connective tissue, umbilical cord and cartilage .



The diagram illustrates the repeating disaccharide unit of hyaluronic acid, enclosed in brackets with a subscript 'n'. It consists of two pyranose rings: a glucuronic acid (GlcUA) unit on the left and an N-acetylglucosamine (GlcNAc) unit on the right. The GlcUA unit is in its chair conformation, featuring a carboxylate group (COO-) at C5, hydroxyl groups at C2 and C3, and a hydroxymethyl group at C6. The GlcNAc unit is also in a chair conformation, with an N-acetyl group (HNC(=O)CH3) at C2, a hydroxyl group at C3, and a hydroxymethyl group at C6. The two units are linked by a beta-1,3-glycosidic bond between the C1 of GlcUA and the C3 of GlcNAc. The labels 'GlcUA' and 'GlcNAc' are positioned below their respective rings, and the linkage is labeled 'β1,3'.

***Read about the functions of Heteropolysaccharides(GAGs), Very important.**

31. Which of the ff is a glycerol based lipid?(Biochemistry)

- A. Phosphatidylcholine/ Lecithin
- B. Ganglioside
- C. Sphingomyelin
- D. Vitamin D
- E. Prostaglandin

ANS. A

32. Which of the ff best describes glucose?(Biochemistry)

- A. It is utilized in biologic systems in L-isomeric form
- B. It's a ketose and usually exists as a furanose ring in solution
- C. It's a C-4 epimer of galactose
- D. It's produced from dietary glucose by the action of alpha-amylase
- E. Homopolysaccharides of Glucose are always branched molecules

ANS. D

33. Most important reason for the special properties of water is

- A. The covalent bonding pattern in water molecule
- B. The bond angle between the two hydrogen atoms
- C. Hydrogen bonding between water molecules
- D. Water can be immediately ionized at room temperature
- E. Water can exist as solid

ANS. C

34. Our immune cells do not attack the self cells in the normal condition. This is because of

- A. Identify membrane proteins on our cells
- B. Identity membrane carbohydrates on our cells
- C. Receptor membrane proteins on our cells
- D. Transporter membrane proteins on our cells
- E. Identify membrane integrins on our cells

ANS. B

35. Which of the ff substance passes through the lipid bilayer by simple diffusion?

- A. Sodium B. Carbohydrate C. Potassium D. Vitamin A E...

ANS. D.....Vitamins A, D, E and K...by simple diffusion through the lipid bilayer yalfalu without the need of protein channels, they are lipid soluble. But Ions and water are lipid insoluble. Thus, pass through water filled channels

36. A type of synovial joint in which a cylindrical shaped projection of one bone fits into a trough shaped surface of another bone with movement occurs in a single plane is (Anatomy)

- A. Plane
- B. Hinge
- C. Condylloid
- D. Saddle
- E. None of the above

ANS. B

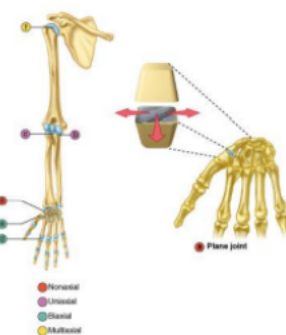
Types of Synovial Joints

2. Hinge joint

- a cylindrical shaped projection of bone fits into a trough shaped surface of another bone
- Motion is within a single plane (**uniaxial**)
- Joint components resemble that of a mechanical hinge

☐ Examples

- Elbow, knee, finger joints



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37. All of the ff cranial nerves carry sensory, motor and parasympathetic fibers EXCEPT

- A. Facial nerve
- B. Oculomotor nerve
- C. Glossopharyngeal nerve
- D. Vagus nerve
- E. None of the above

ANS. B.....because Oculomotor is not sensory

***Betechemari..Cranial nerves III, VII, IX and X and Sacral nervs 2, 3 and 4 carry parasympathetic fibers.**

38. Wrong about the process of cross over (Cytology)

- A. A chiasma is formed at the point of interchange of chromatid segments of homologous chromosomes.
- B. It occurs at meiotic II
- C. Results genetic variability
- D. Random distribution of homologous chromosomes to daughter cells
- E. None of the above

ANS. B.....Crossover happens at Prophase I, first stage of meiosis I

39. A unique type of joint between a tooth and its socket (Anatomy)

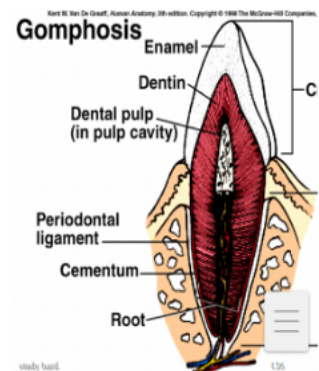
- A. Suture
- B. Syndesmoses
- C. Gomphoses
- D. Symphyses
- E. None of these

ANS. C

GOMPHOSES

➤ unique joint b/n a tooth & its socket

➤ **Fibrous tissue** holds teeth in their sockets



40. What is the osmotic pressure of a solution with 150 mmol/L solution of Calcium Chloride(CaCl_2)?(Physiology)

- A. 5,798mmHg B. 450mmHg C. 38.6mmHg D. 8,685mmHg E. 6,500mmHg

ANS. D

Osmotic pressure = ?

Osmotic pressure = osmolarity * 19.3mmHg

Osmolarity = Concentration * number of dissociable ions

Concentration = 150mmol/L

Number of dissociable ions = 3,..... 3 because when Calcium Chloride(CaCl_2) dissolves in water it dissociates into 1 Calcium and 2 Chloride ions.

-Osmolarity = $150 * 3 = 450$

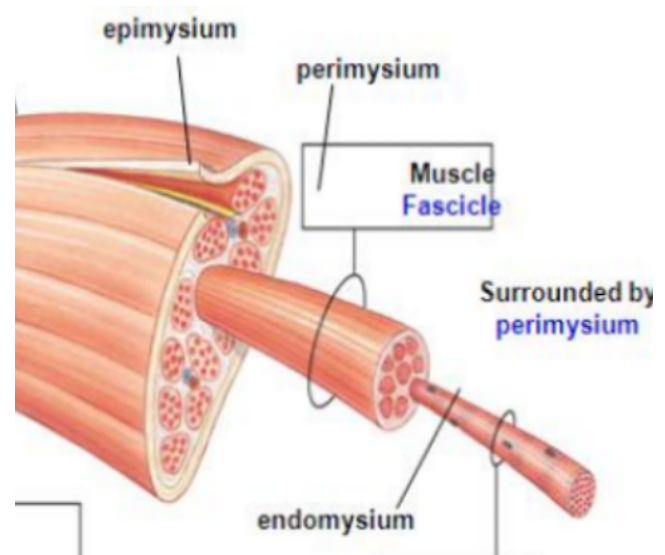
-Osmotic pressure = $450 * 19.3 = 8,685\text{mmHg}$

* But if the concentration was given in g/L, don't forget to convert it to mol/L by dividing it by the molar mass of the compound (In this case CaCl_2)

41. The ff are connective tissue layers around the skeletal muscle. Identify the correct order from internal to external (Anatomy)

- A. Endomysium...Perimysium...Epimysium
 B. Perimysium...Endomysium...Epimysium
 C. Endomysium...Epimysium...Perimysium
 D. Epimysium...Perimysium...Endomysium

ANS. A



42. Which of the following is a commonly used clearing agent in tissue section preparation for light microscopy?(Histology)

- A. Ethanol B. Methanol C. Paraffin wax D. Formalin E. Xylene

ANS. E

43. In hematoxylin and eosin method of the general tissue section staining, hematoxylin (Histology)

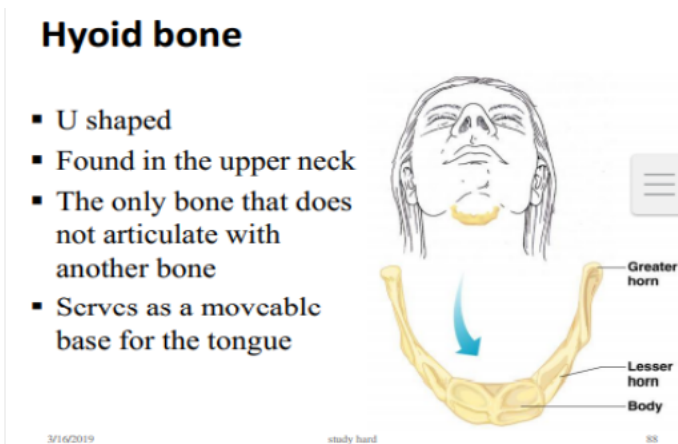
- A. Acts like an acidic dye C. Stains blue black E. B and C
B. Stains best the ribosomes D. A and B

ANS. E

44. The only bone that doesn't articulate with any other bone is (Anatomy)

- A. Sternum
B. Nasal bone
C. Vomer
D. Hyoid bone
E. None of these

ANS. D



45. Which one of the following is the correct order of tissue block preparation from an autopsy tissue? (Histology)

- A. Dehydration, fixation, clearing, infiltration, & embedding
B. Dehydration, clearing, infiltration, fixation & embedding

- C. Fixation, dehydration, clearing, infiltration, & embedding
- D. Fixation, dehydration, infiltration, clearing & embedding
- E. Clearing, dehydration, fixation, infiltration, & embedding

ANS. C (Mnemonics - Find DC In East)

46. The membrane of which of the following organelle is continuous with the outer nuclear (Cytology)

- A. Centrioles
- B. Microtubule
- C. Golgi apparatus
- D. Rough endoplasmic reticulum
- E. Smooth endoplasmic reticulum

ANS. D

48. Which one is False about Anatomical Position?(Anatomy)

- A. Body erect
- B. Head, eyes, and toes directed forward
- C. Feet together
- D. Arms at side with palms facing backward
- E. None of the above

ANS. D.....The palms should face forward, not backward.

49. Vesicles produced from rough endoplasmic reticulum will be transported to which organelle?(Cytology)

- A. Smooth endoplasmic reticulum
- B. Lysosome
- C. Peroxisome
- D. Golgi apparatus
- E. Microtubule

ANS. D

50. Which one of the following movement is different from the others?(Anatomy)

- A. Protrusion
- B. Flexion
- C. Extension
- D. Abduction
- E. Hyperextension

ANS. A.....Protrusion is a special type of movement, while the rest are Angular.

51. Which one of the following bone markings is not related to joints?

- A. Head
- B. Facet
- C. Condyle
- D. Foramen
- E. None of these

ANS. D.....Foramen is an opening through a bone that allows bloodvessels and nerves to pass

TABLE 5.1 Bone Markings (continued)	
Name of bone marking	Description
Projections That Help to Form Joints	
Head	Bony expansion carried on a narrow neck
Facet	Smooth, nearly flat articular surface
Condyle (kon'dil)	Rounded articular projection
Ramus (ra'mus)	Armlike bar of bone
Depressions and Openings Allowing Blood Vessels and Nerves to Pass	
Meatus (me-a'tus)	Canal-like passageway
Sinus	Cavity within a bone, filled with air and lined with mucous membrane
Fossa (fos'ah)	Shallow, basinlike depression in a bone, often serving as an articular surface
Groove	Furrow
Fissure	Narrow, slitlike opening
Foramen (fo-ra'men)	Round or oval opening through a bone



52. All of the following statements are true about sesamoid bones except:

- A. Prevent excessive wear and tear
- B. Helps tendons to glide over bony surfaces
- C. Alter the direction of pull of the tendons
- D. Decrease mechanical advantage in the joints of long bones
- E. None of the above

ANS. D,.....Sesamoid bones increase mechanical advantages

53. Which one is a correct order in structural organization of skeletal muscle from cell to the whole muscle?

- A. Fascicle – muscle fiber – muscle
- B. Muscle fiber – fascicle – muscle
- C. Muscle – muscle fiber – fascicle
- D. Muscle fiber – muscle – fascicle
- E. None of the above

ANS. B

54. The amino acid which can serve as proton donor and acceptor is:

- A. Glycine B. Proline C. Histidine D. Isoleucine E. Tryptophan

ANS. C,....., Histidine is 10% protonated at pH of 6, so it can act both as an acid and base.

55. A semi-essential amino acid that is required especially during pregnancy is (Biochemistry)

- A. Methionine B. Valine C. Arginine D. Leucine E. Tyrosine

ANS. C

56. Which of the following is not a function of smooth endoplasmic reticulum? (Cytology)

- A. Synthesis of steroid hormones
B. Detoxification
C. Storage and release of calcium ions in the striated muscle cells
D. All of the above
E. None of the above

ANS. E

57. Which of the following is wrong about microtubules?(Cytology)

- A. Are made from 13 protofilaments arranged circumferentially
B. Could increase in length but not in width
C. Transport vesicles within their luminal space from one region to another region in the cell
D. All of the above
E. None of the above

ANS. C

Cytoskeleton

- Structural framework within the cytosol.
- Functions: maintain cell shape, stabilize cell attachments, facilitate endocytosis and exocytosis, and promote cell motility.

Microtubules

- are straight, hollow tubules 24-25 nm in diameter.
- have a rigid wall composed of 13 protofilaments, each of which consists of a linear arrangement of $\alpha\beta$ -tubulin dimers.
- are polar, with polymerization (assembly) and depolymerization (disassembly) occurring preferentially at one end (+ end).
- contain microtubule-associated proteins (MAPs), which stabilize them and bind them to other cytoskeletal components and organelles.
- are associated with *kinesin*, a force-generating protein, which serves as a "motor" for vesicle or organelle movement.

Functions

- maintain cell shape
- aid in the transport of macromolecules within the cytosol (axoplasmic transport, melanin transport, vesicle movements b/n ER & GC and GC & membs)
- promote the movement of cilia, flagella, and chromosomes

Paclitaxel, vinblastine and colchicine are good examples of microtubule interferers/

58. Regarding the nucleolus, which of the following statements is true?(Cytology)

- A. Is acidophilic
- B. Is the site of mitochondrial synthesis
- C. Is membrane bound region within the nucleus
- D. Is made by regions of chromatin contributed by different chromosomes
- E. Is not visible under the light microscope

ANS. D,.....Nucleolus is a well-defined, but not membrane-bounded nuclear inclusion. It is usually basophilic when stained with hematoxylin and eosin.

59. Organelles made by stalk of 3-10 discrete flattened and slightly curved bag-like channels or cisternae are (Cytology)

- A. Mitochondria
- C. Peroxisome
- E. Microtubules
- B. Smooth endoplasmic reticulum
- D. Golgi bodies

ANS. D

60. Which nerve doesn't contain parasympathetic fibers?

- A. Trigeminal
- B. Vagus
- C. Glossopharyngeal
- D. Oculomotor
- E. Facial

ANS. A,.....Cranial nerves III(Oculomotor), VII(Facial), IX(Glossopharyngeal), X(Vagus) and Sacral nerves 2, 3 & 4 contain parasympathetic nerve fibers while the rest contain sympathetic.

61. During the cell cycle, (Cytology)

- A. Duplication of centrioles begins and completes within the G1 phase
- B. The S phase is the longest and most variable phase
- C. Daughter cells grow to the size of the parent cell during G1 phase
- D. Synthesis of RNA and proteins occurs mainly during the S phase
- E. Replications of the chromosomes occurs during G2 phase membrane?

ANS. C

CELL CYCLE

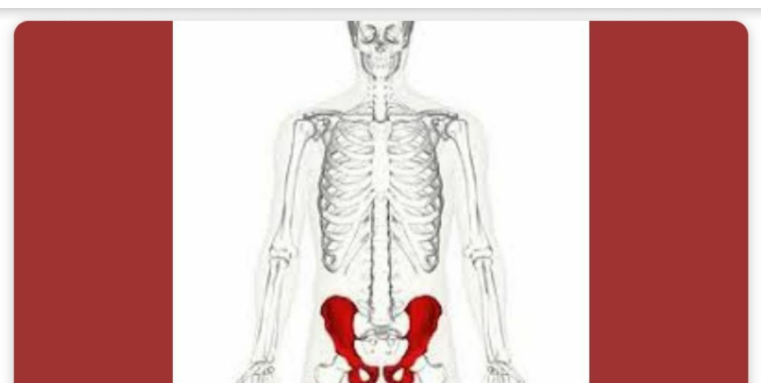
- * Sequence of events that is executed repetitively; repeated divisions at regular intervals
- * Dynamic & continuous process,
- * Induced by phytohemagglutinin (NGF, EGF, FGF, precursors of RBC GF=erythropoietin)
- * Arrested in metaphase stage by colchicine (chalones)
- * Consists of two major periods
 - Interphase; interval between cell divisions,
 - Mitosis; periods of cell division (M phase),
- ⇒ Interphase;
 - Considerably longer than the M phase
 - Cell doubles in size and DNA content
- * Phases
 - G₁ Phase , Cell growth & protein synthesis (restore daughter cells to normal vol. & size)
 - Restriction point - "trigger proteins" synthesized to reach a threshold,
 - Cells that fail to reach the restriction point = resting cells, enter G₀ phase,
 - Lasts from few hours to several days,
 - S Phase , DNA replication & protein synthesis = duplication of chromosomes
 - Centrioles self-duplicate
 - Lasts 8-12 hours in most cells,
 - G₂ Phase , Cell prepares to divide
 - Centrioles grow to maturity
 - Energy required for the completion of mitosis is restored
 - RNA & proteins necessary for mitosis are synthesized
 - Lasts 2-4 hours,
- * Control factors
 - S-phase activator triggers the initiation of DNA synthesis
 - M-phase delaying factor
 - M-phase promoting factor

62. The three hipbones; ilium, ischium and pubis join or merge at a site called (Anatomy)

- A. Acetabulum B..... C..... D..... E. None of the above

ANS. A, Acetabulum

In humans, the first elements to fuse are the ischium and pubis, which unite anteriorly to form the ischiopubic ramus between 4 and 8 years of age. Next, the ilium fuses to the combined ischiopubic portion at the acetabulum between 11 and 15 years in females and 14 to 17 years in males to form the os coxa.



The **hip bone** is formed by **three** parts: ilium, ischium, and pubis. At birth, these **three** components are separated by hyaline cartilage. They join each other in a Y-shaped portion of cartilage in the acetabulum. By the end of puberty the **three** regions will have **fused together**, and by the age 25 they will have ossified.

63. Oocyte like ovary endometrium develop and renew layer yet grow new?(Embryology)

A. Zona pellucida B. Acrosome C..... D..... E....

ANS. Not sure, couldn't recall the other choices, find out by urself!

64. There was a question about hydrostatic pressure and osmotic pressure, the hypertonic solution gave rise to water, study them. (Physiology)

65. Which layer the uterus disintegrates and is shed off during menstruation? (Embryology)

A. Basal layer of endometrium C. Perimetrium
B. Functional layer of endometrium D. Myometrium E....

ANS. B,.....The functional layer, composed of compact and spongy layers, disintegrates and is shed during menstruation and after parturition (delivery of a baby).

66. Which of the following mostly store glycogen?(Biochemistry)

A. Kidney and Muscle C. Liver and Muscle
B. Kidney and Liver D..... E.....

ANS. C

67. Which of the following are multinucleated cells?

A. Megakaryocytes B. Monocytes C. Osteocytes D.... E...

ANS. C,.....Osteocytes are multinucleated bone cells, Megakaryocytes(have a large nucleus)& Monocytes are mononucleated.

68. Which of the ff nerve is Not connected to the hind brain? Yemil question neber.

ANS.

2 Cranial nerves, I(Olfactory) & II(Optic) emerge from the forebrain

2 Cranial nerves, III(Oculomotor) & IV(Trochlear) emerge from the midbrain

8 Cranial nerves; V(Trigeminal), VI(Abducent), VII(Facial), VIII(Vestibulocochlear), IX(Glossopharyngeal), X(Vagus), XI(Accessory) and XII(Hypoglossal) orginate from the hindbrain.

*We can also say.....cranial nerves I & II emerge from fore brain, while the rest 10 nerves(III-XII) emrge from Brain stem.

69. Which one of the ff is not an artery or vein found in the upper limb?

A. Subclavian vein/ artery

D. Cephalic vein

B. Axillary artery

E. Basilic vein

C. Brachial artery

ANS. A

70. All of the ff amino acids are optically active except(Biochemistry)

A. Glycine

C.....

B.....

D.....

E...

ANS. A,.....Glycine is optically Inactive

glycine

The simplest member of this group is **glycine**, where the saturated carbon atom is **unsubstituted**, rendering it optically inactive. The rest of the 20 most common amino acids are optically active existing as both D and L stereoisomers.

71. When 10 glucose molecules ($C_6H_{12}O_6$) through dehydration reactions sidemeru yetegnawun yesetalu? (Biochemistry)

- A. $C_{60}H_{120}O_{60}$ B. $C_{60}H_{120}O_{51}$ C. $C_{60}H_{102}O_{51}$ D. $C_{60}H_{102}O_{60}$

ANS. C



....endi eyale 10 glucose molecules through 9 reactions will form an oligosaccharide, in each step a molecule of water eyeweta malet new. Which means 2 hydrogen atoms and 1 oxygen atom eyewetu new but no carbon atom is lost.

So,

-we have 6 carbon atoms in a glucose molecule..... $6 \times 10 = 60$

C_{60} ...

-We have 12 hydrogen atoms in a glucose molecule..... $12 \times 10 = 120$

But in each of the 9 reactions 2 hydrogen atoms are lost which means throughout the 9 reactions $2 \times 9 = 18$ hydrogen atoms are lost

$$120 - 18 = 102$$

$C_{60}H_{102}$...

-We have 6 oxygen atoms in a glucose molecule..... $6 \times 10 = 60$

But in each of the 9 steps an oxygen atom is lost which means throughout the 9 reactions $1 \times 9 = 9$ oxygen atoms are lost

$$60 - 9 = 51$$

$C_{60}H_{102}O_{51}$This is our answer!

72. Which one of the ff is true? (cytology)

A. Mitochondria has its own DNA which is circular

B.....

C.....

D.....

E.....

ANS. A,....couldn't remember the other choices.

73. About Passive process across cell membrane question Meto neber, passive transport yehonev neber milew (physiology)

-efflux, influx

-efflux of sodium, of potassium

ANS. The answer is Efflux of potassium!

* To answer this question, under normal conditions more intracellularly than extracellularly more or less

-Under normal conditions; Sodium, Calcium & Glucose are more concentrated extracellularly While potassium, phosphate & Sulfates are more concentrated intracellularly. As u know passive diffusion is from high concentration to low concentration.

-Influx means into the cell move, efflux move out of cell into the extracellular matrix

* For example:

-Efflux of potassium/sulphates/phosphates as well as Influx of sodium/glucose/calcium follow passive type of diffusion.

But

-Influx of potassium/sulfates/phosphates as well as efflux of sodium/glucose/calcium use active transport.

74. Which one of the ff is true?(Cytology)

A. microtubules are more stable than microfilaments

B....

C...

D....

E...

* Couldn't remember the other choices. Bcha A is NOT true, it's the other way around (Microfilaments are more stable than microtubules), maybe the question 72 has the same question like yechlal.

75. Which one is NOT an angular movement?(Anatomy)

A. Moving the trunk to the thigh

B. Moving the forearm towards the arm

C. Moving the head to the side

D....

E....

ANS. Couldn't remember the other choices but A, B & C are Angular movements

76. Which tarsal bone makes an articulation with both tibia and fibula?(Anatomy)

A. Calcaneal

B. Talus

C. Cuboid

D. Navicular

E....

ANS. B

Name the tarsal bone that articulates with the tibia and fibula to form the ankle joint.

talus

77. Which of the ff is an example of 20 ATP Active transport?(Physiology)

(Take care ezi question lay 20 mitlewa tesasta new secondary'n be symbol lemetsaf tefelgo neber.....)

- A. Na-K pump
- B. Na-Glucose pump
- C..... D..... E.....

Two types active transport

A) Ion driven

➤ Down hill ion movements uphill the solutes

➤ Indirect source of energy is ATP

B) ATP-driven: primary or direct

➤ ATP hydrolyzed by ATPase to drive uphill solute

*Look in the picture there are two types of active transport

-Ion driven and ATP driven

The thing is ATP driven directly ATP'n as a source yemitekem transport selehone it's called primary Active transport,

ION driven'u demo it's called secondary transport.

*So, the answer is most probably B(Na-Glucose pump), an Ion driven/secondary active transport, in which down hill movement of ions/Na/ results in an uphill movement of Solutes/Glucose/

78. During which embryonic month do the primary oocytes enter meiosis I and get arrested in prophase?(Embryology)

- A. During the seventh month
- B. Beginning in the third month
- C..... D..... E.....

ANS. B

79. Which of the ff is true?(Embryology)

A. NO primary oocytes are formed after birth.

B..... C..... D..... E.....

ANS. A,.....hulum before birth new tesertew miyalkut

80. Which one is true about Non gated channels?(Physiology)

A. Sodium excess minamin

B. They decrease concentration gradient

C..... D..... E.....

ANS. B

81. Which is true about tonicity?(Physiology)

A. When ECC(Extracellular compartment) and ICC(Intracellular compartment) have equal concentration, there is no movement of fluid.

B..... C..... D..... E.....

ANS. A,...couldn't recall the other choices.

82. There was a question about nucleoli, Mitochondria cristae minamin(cytology)

And another question about facilitated diffusion.(Physiology)

Know this: rRNA mimeretew it's inside the nucleolus and cristae mibalut they are foldings in the inner mitochondrial mnamn...

83. Which joint is formed between diaphysis and ephiphysis of a long

bone?(Anatomy)

A. Symphyses

B. Syndesmoses

C. Suture

D. Synchondroses

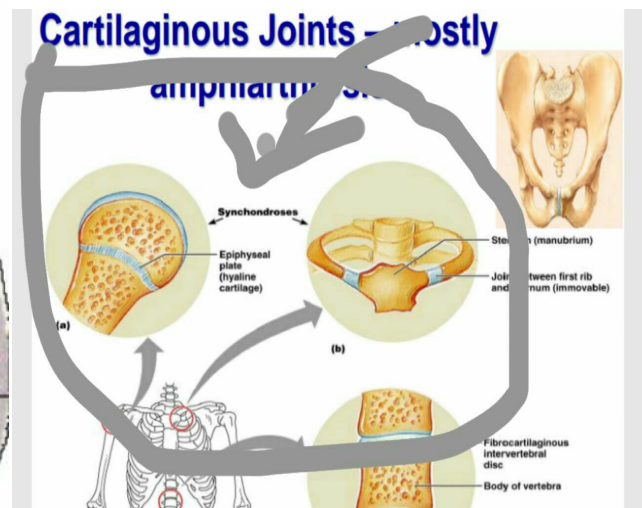
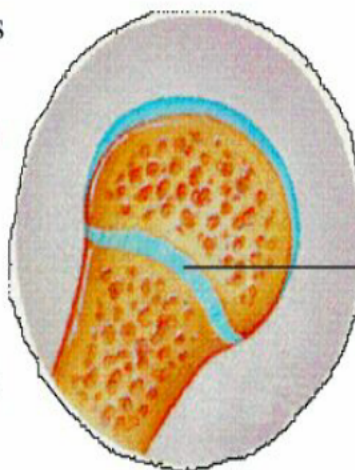
E. Gomphoses

ANS. D

There's the ephiphyseal line/plate in between the diaphysis and ephiphysis so

SYNCHONDROSES

- Hyaline cartilage unites the bones
- E.g.. Epiphyseal plates in growing children
- Provide for bone growth
- When growth ends all synchondroses become immovable



zoom

84. Which buffer yishalal yemil tiyake neber....(couldn't recall clearly).....(physiology)

*Well le extracellular kehone carbonate buffer is very konjo

*For intracellular kehone demo phosphate buffer

85. There was yehone sewye betam fat rich food mibela hul gize, keza ye fat amount in his body reached 350mg/dL but normal amount of fat in the body should be less than 150mg/dL. Keza the sewye goes to see a doctor and the doctor addis ye amegageb style (low fat, low protein and high carbohydrate foods) endiketel yenegrewal, then sewyew yehen ye amegageb style teketilo, after some time endegena fat amount in his body measure sidereg yebas wede 700mg/dL chemerual? What could have caused this increase in fat level?

- A. Low intake of protein
- B. Low intake of fat
- C. High intake of carbohydrate
- D. Phospholipid.....
- E.....

ANS. C

86. Which is true about the chambers of the hearttt

Mircha wust..

- apex minamin, right ventricle constitutes most of the heart, left atrium is the largest structure superiorly minamin.....

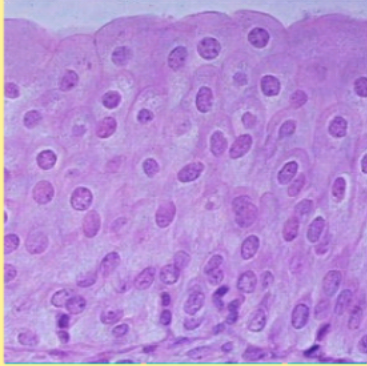
-To answer this read about The Cardiovascular system, which is one part of the circulatory system. The other part is The Lymphatic system, there was one question about this system.

REGARDING HISTOLOGY

Focus on these they're from Dr. Girmai's Slide

Examples: Epithelial tissue

- Transitional epithelium: (Urothelium) found in the bladder
- Stretches as the bladder becomes full



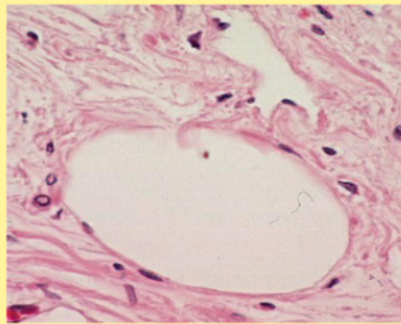
Examples: Epithelial tissue

- Stratified squamous: lines the mouth, esophagus, cervix, skin
- Several layers offers protection to outer layers and membranes of the body.



Examples: Epithelial tissue

- Simple squamous: lines blood vessels, lungs...
- Allows
 - increased blood flow
 - increased oxygen diffusion



Examples: Epithelial tissue

- Simple columnar digestive tracts...
- Cells mixed with *goblet cells* that secrete mucus



* Slenezi types of tissues read bedemb, their functions, which organ protect endemiyaregu, yet endemigegnu (especially transitional epithelium, found in the bladder).....focus

2. Histotechnique: Preparation of Tissues for Microscopic Examination

- ⇒  **Fixation:** Chemical, [Immersion, Perfusion]; Thermal; Frozen Section;
- ⇒  **Processing:** Dehydration, Clearing and Infiltration
- ⇒  **Embedding:** agar, gelatine or wax, epoxy resins
- ⇒  **Sectioning:** Microtomy (microtome), Cryomicrotomy (cryostat)
- ⇒  **Staining:** Hematoxylin & eosin (LM), Uranyl acetate and lead citrate (EM).

⇒  **Special Staining:** safranin, oil red O, Congo red, fast green, silver salts, and numerous natural and artificial dyes developed for the textile industry.

* Enezin steps, min min endemifeter in each steps

- * Cells are difficult to see by LM. To help us visualize the structure and features of cells, dyes are used to impart a particular color to cells. These dyes react with different chemical features of proteins, nucleic acids and carbohydrates and can be used to highlight certain cellular structures.
- * **Hematoxylin and eosin (H&E)** → standard method of staining. * Hematoxylin is a basic dye (positively charged) that binds to negatively charged DNA and RNA and is blue in color.
- * Eosin is pink in color and is an acidic dye (negatively charged) that binds to positively charged particles like mitochondria. Positively charged structures are therefore said to be "eosinophilic."
- * While H&E staining is widely used, it is limited in its ability to differentiate between cytoplasmic organelles and many tissue components.
- * **Periodic acid-Schiff method (PAS)** is useful for staining structures rich in polysaccharides (glycogen) mucopolysaccharides (ground substance, basement membrane, mucus) glycoproteins (thyroglobulin), and glycolipids.
- * periodic acid oxidizes 1,2-glycols and 2,2-amino alcohols to aldehydes, which are then stained reddish purple by Schiff reagent.
- * **Osmium staining** blackens lipids and stains the Golgi apparatus under LM. It is also used as a fixative for electron microscopy.

*Echin page(34) bedemb...everything.....፬፬፬፬...

*Histo eske page 40 deres manebab bedemb, keza behuala yalew part not that much important, fetenam lay almetam neber.

GOOD LUCK!!!

